

Describing Data

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **distinguish** categorical and quantitative data.

RELATED SKILLS

- I Classify data as categorical or quantitative.

- ☐ 2. I can **construct and interpret** frequency and relative-frequency tables.

RELATED SKILLS

- I Calculate and interpret relative frequency.
- I Construct and interpret a frequency table.

- ☐ 3. I can **construct and interpret** bar charts, pie charts, and histograms using appropriate technology.

RELATED SKILLS

- I Construct and interpret a histogram for quantitative data.
- I Construct and interpret bar charts and pie charts.

- ☐ 4. I can **identify** misleading features in a data display.

RELATED SKILLS

- I Identify perceptual distortion, misleading scales, and other misleading features in data displays.

- ☐ 5. I can **describe** a distribution's modality, symmetry, skewness, and outliers.

RELATED SKILLS

- I Describe a distribution by modality, symmetry, and skewness.
- I Identify and interpret unusual observations or outliers.
- I Recall and correctly use terms for data type, distribution shape, center, spread, percentile, and outlier.

- ☐ 6. I can **calculate and interpret** mean, median, and mode and select an appropriate measure of center.

RELATED SKILLS

- I Calculate and interpret the arithmetic mean.
- I Calculate and interpret the median.
- I Identify and interpret the mode or modes.
- I Select an appropriate measure of center based on distribution shape and context.

☐ **7. I can **calculate and interpret** range, standard deviation, quartiles, and interquartile range.**

RELATED SKILLS

- ☐ Calculate and interpret standard deviation as a measure of spread.
 - ☐ Calculate and interpret the range.
 - ☐ Determine quartiles and calculate and interpret the interquartile range.
-

☐ **8. I can **construct and interpret** a five-number summary and boxplot.**

RELATED SKILLS

- ☐ Construct and interpret a boxplot.
 - ☐ Determine and interpret a five-number summary.
-

☐ **9. I can **use** percentiles and z-scores to locate and compare values within distributions.**

RELATED SKILLS

- ☐ Calculate and interpret a z-score.
 - ☐ Determine and interpret the percentile rank of a value.
-

☐ **10. I can **compare** distributions using their shape, center, spread, and unusual features.**

RELATED SKILLS

- ☐ Compare distributions using shape, center, spread, and unusual features.
 - ☐ Describe a distribution by modality, symmetry, and skewness.
 - ☐ Select an appropriate measure of center based on distribution shape and context.
-